



NIRAJ NAIDU

MECHANICAL DESIGN ENGINEER

+49 15757055980 nirajnaidu0411@gmail.com 11 April 2001 www.linkedin.com/in/nirajnaidu11 Essen, 45138

A passionate and detail-oriented Mechanical Design Engineer with practical experience in 3D modeling, product design, and manufacturing processes. Skilled in SolidWorks, CATIA, and ANSYS, with a strong foundation in hydraulic system design, reverse engineering, and precision machining. Experienced in leading Formula Student projects, designing EV components, and developing industrial equipment such as gantry cranes, mold presses, and tube bending machines. I combine creativity with technical expertise to deliver accurate, efficient, and innovative mechanical solutions—whether in professional engineering environments or independent freelance projects.

EDUCATION

M.Tech Digital Technologies

FOM University of Applied Sciences
for Economics and Management,
Germany
2024–Present

B.Tech Mechanical (7 CGPA of 10)

Symbiosis Institute of Technology,
Pune
2021–2024

12th Science – 70.77%

Fr. Agnels Multipurpose School and
Jr. College
2017–2018

10th – 75.00%

Fr. Agnels Multipurpose School and
Jr. College
2015–2016

SKILLS

- SOLIDWORKS
- CATIA
- AUTOCAD
- CURA
- RD WORKS
- ANSYS
- 3D PRINTING

LANGUAGES

- English
- Hindi
- Marathi
- German



PROJECT AND WORK EXPERIENCE

KC Engineering and Automation

Design Engineer (1st June 2024 – 31st December 2024)

- Designed and modeled hydraulic systems such as cylinders, presses, and lifting equipment using SolidWorks.
- Executed reverse engineering of the SLB 1100 Gantry Crane and supported the development of tube bending and mold press machines.
- Prepared detailed CAD drawings and ensured design precision, manufacturability, and assembly accuracy.

ASB International Pvt. Ltd.

Design Intern (5th January 2024 – 10th May 2024)

- Created 3D models of mold components and designed precision fixtures for industrial machinery.
- Maintained tolerances up to 5 microns to ensure assembly stability during machining.
- Supported the design team in drafting and documentation for production processes.

Freelancing

3D Printing Product and Design (1st January 2024)

- Deliver 3D modeling and 3D printing solutions for clients, ensuring high accuracy and design efficiency.
- Develop prototypes and production-ready CAD models tailored to client specifications.
- Provide design consulting focused on precision, material selection, and assembly fit.

Wrench Welders Racing – Formula Student Team (SIT)

Design Head (2022–2023)

- Executed the reverse engineering of the SLB 1100 Gantry Crane to improve structural efficiency.
- Designed and developed hydraulic tube bending machines, mold press equipment, and horizontal moving jacks.
- Created detailed 3D CAD models ensuring precision and manufacturability.
- Contributed to the development of custom hydraulic systems for industrial applications.

KC Engineering and Automation – Industrial Projects

Design Engineer (1st June 2024 – 31st December 2024)

- Led the design and fabrication of Formula Student race car components using SolidWorks and CATIA.
- Worked on carbon fiber and fiberglass bodywork, and developed a high-efficiency EV battery system that ranked 15th among 72 teams.
- Developed an engine sound signature segregation system using audio analysis and machine learning for diagnostics.
- Represented the team at major competitions including Supra 2022 and Formula Student EV 2021